CHAPTER

ATA 45

CENTRAL DIAGNOSTIC SYSTEM

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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CONTENTS

CENTRAL DIAGNOSTIC SYSTEM, GENERAL

Introduction

The Central Diagnostic System (CDS) in the on−board maintenance system of the Dash 8−Q400. It provides a maintenance function that uses the diagnostic features of the installed avionics systems (IMS 100 Avionics System). The CDS gives information displays and control from the Audio and Radio Control Display Unit (ARCDU) which is used as the CDS interface with maintenance personnel.

The CDS supplies:

- Diagnosis of the avionics systems, which includes information display and control from the flight compartment
- Line Replaceable Unit (LRU) tests and failure reports
- A display of the information stored in the aircraft maintenance module.

General Description

Refer Figure - Central Diagnostic System Block Diagram
Refer Figure - CDS Architecture

The central maintenance system has the sub−systems that follow:

- 45−45−00 Central Diagnostic System
- 45−61−00 Propeller Balance Monitoring System.

Centralized Diagnostic System and System Architecture

The Centralized Diagnostic System (CDS) is used to monitor, control, record and show the system tests, diagnostics and maintenance activities.

The CDS lets aircraft maintenance personnel to do the procedures that follow:

- Read the avionics status report
- Read malfunction reports from last or previous flight legs
- Do fault isolation and return to service tests after maintenance is completed
- Read the part number of Line Replaceable Units (LRU) it monitors.

NOTE

Note: The CDS does not access the Built In Test Equipment (BITE) of the avionics systems it interfaces with. Special test equipment during shop maintenance is used to make this troubleshooting data available.

The pilot system checks to make sure that the systems function correctly before flight is done by related switches in the flight compartment and not by the CDS.

The system architecture is in four categories:

- CDS storage capacity
- Failures classification
- CDS architecture
- CDS/FDPS interface.

CDS Storage Capacity − Failure messages are stored in the CDS non−volatile memory (NVM) with the storage capacity that follow:

- 700 failure messages in the Flight Report Zone (Last Leg report zone + Previous Legs report zone) with the repartition as follows: Maximum 50 failure messages in the Last Leg report zone, the remaining storage capacity is reserved for the Previous Legs report zone.

- 100 failure messages in the MAINT zone.

Failures Classification − Failures are classified as follows:

- Class 1 failure is a failure of a system or a LRU that prevents the continued operation of the associated system. This failure is associated with a flight compartment effect or indication
- Class 2 failure is a failure of a system or a LRU that allows the continued system operation, but at a reduced capacity level, such as a loss of redundancy. Flight compartment effects are not associated with these failures.

CDS Architecture − The CDS is found in the Input Output Processor module 1 (IOP1) inside the Integrated Flight Cabinet No. 1 (IFC1). The avionics equipment which interface with the CDS are classified into three categories:

- Type 1 components (EIS, IFC, RCAU, ARCDU) can dialog through a bidirectional ARINC 429 bus to provide the CDS with their fault reports in real time or upon CDS request (send faults stored in memory or initiate a dedicated test on the system)
- Type 2 components are the ADU and the AHRU. These components can send in real time their fault reports to the CDS and to initiate a test upon CDS request.
- Type 3 components (common hardware) send only their healthy/not healthy status to the CDS through discrete logic signals.

NOTE

The CDS is also used to display maintenance information related to some non−avionics systems: Engine Monitoring Unit, Anti−skid, Flight Control Unit, Ice and Rain Protection System, Timer Control Unit 1 and 2, Cabin Pressure Controller, APU and Environmental Control System.

The access to the CDS can only be done through a maintenance switch with the aircraft in a weight−on−wheels configuration.

CDS/FDPS Interface − The CDS is hosted in the IOP and shares the processor with the Flight Data Processing System (FDPS). The two applications are separate in time and in space.

The CDS can operate in Normal, Maintenance or Downloading Modes:

- In Normal Mode, the CDS is inactive; this is the mode where systems report their fault reports
- In Maintenance Mode, the CDS is active and the interface with the operator is available through the use of one of the ARCDUs

The remote maintenance switch on the Maintenance Panel on the wardrobe shelf is used to get access to the maintenance mode.

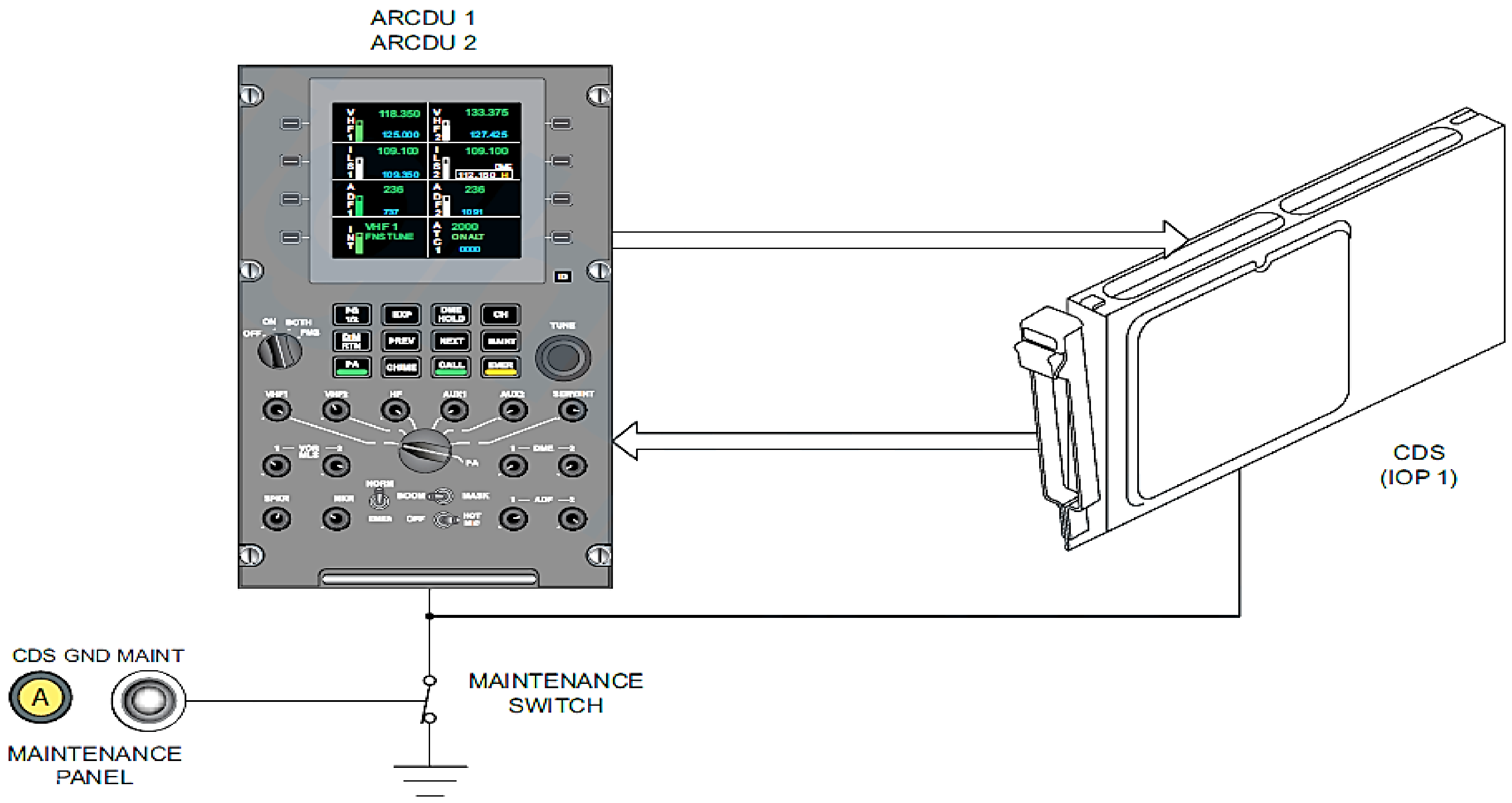


FIGURE - CENTRAL DIAGNOSTIC SYSTEM BLOCK DIAGRAM

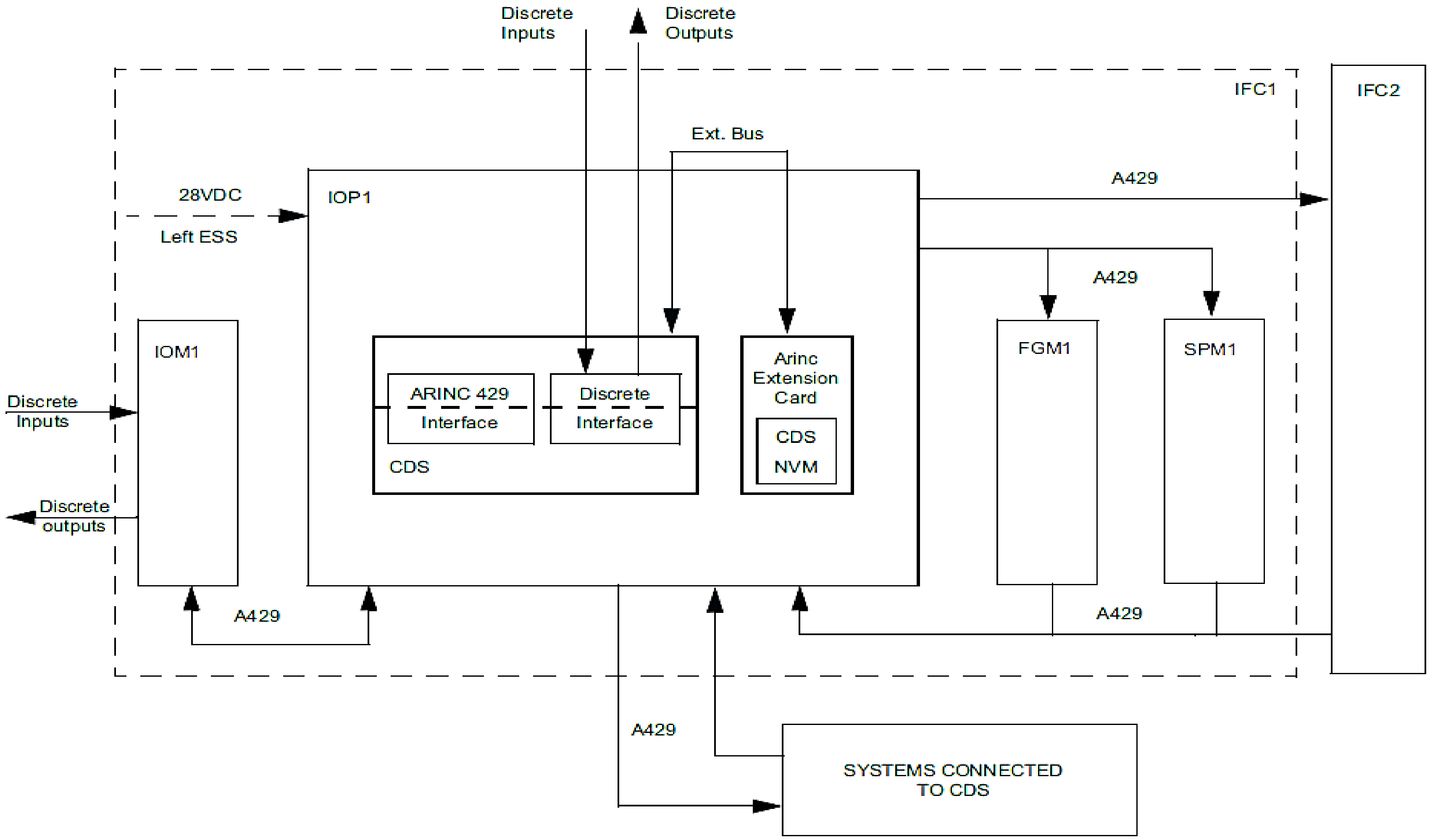


FIGURE - CDS ARCHITECTURE

MAINTENANCE PANEL

Introduction

The maintenance panel is a component of the Central Diagnostic System (CDS). It is used by aircraft maintenance personnel to do the functions that follow:

- Erase fault data
- Troubleshooting at the Line Replaceable Unit (LRU) level
- Calibration tests
- Download fault data.

NOTE

Do not use the CDS as the only source to find faults.

General Description

Refer Figure - CDS Maintenance Panel
Refer Figure - CDS Maintenance Panel Block Diagram

The maintenance panel is installed forward of the passenger door above the wardrobe. The wardrobe cover must be removed to use the maintenance panel. The CDS GND MAINT toggle switch on the maintenance panel must be set to the “up” position to energize the panel.

The maintenance panel has three sections as follows:

- Engine Maintenance Panel
- Systems Maintenance Panel
- System Download Panel.

The maintenance panel interfaces with the systems that follow:

- Powerplant
- Propeller
- Environmental Control System
- Cabin Pressure Control
- Active Noise Vibration Suppression
- Ice and Rain Protection System – Deice
- Propeller
- Airframe (AFR)
- Flight Controls
- Flaps
- Landing Gear
- Auxiliary Power Unit (APU)
- Public Address System.

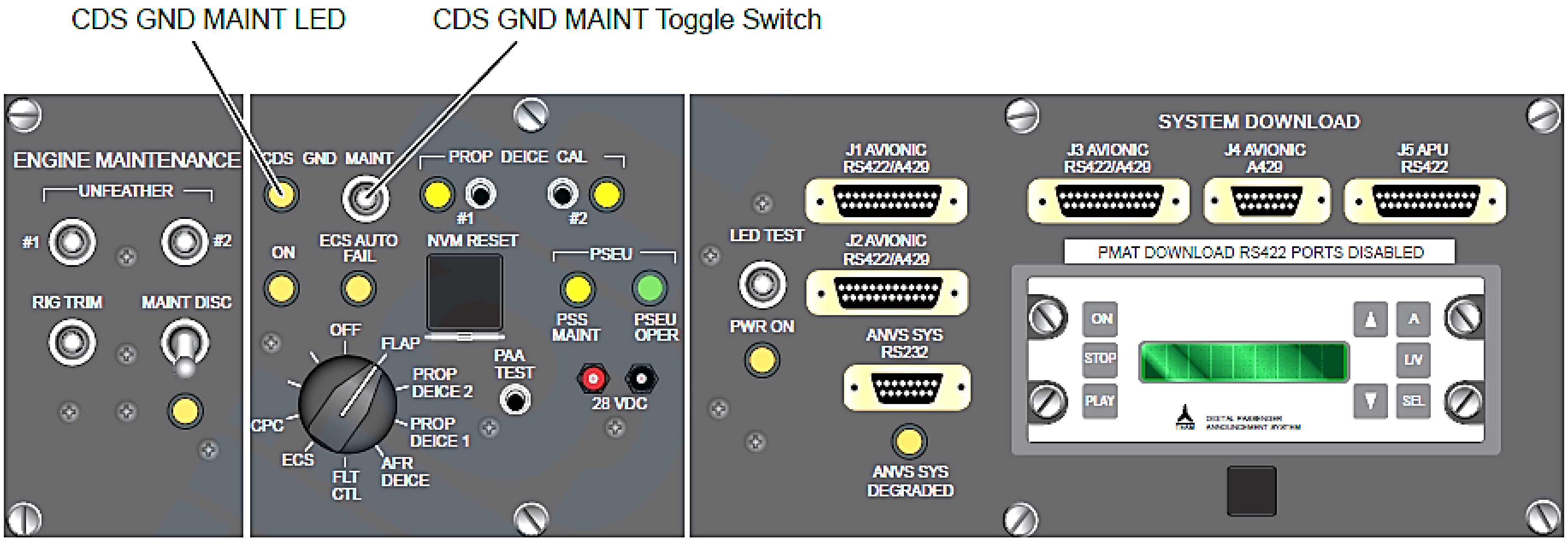


FIGURE - CDS MAINTENANCE PANEL

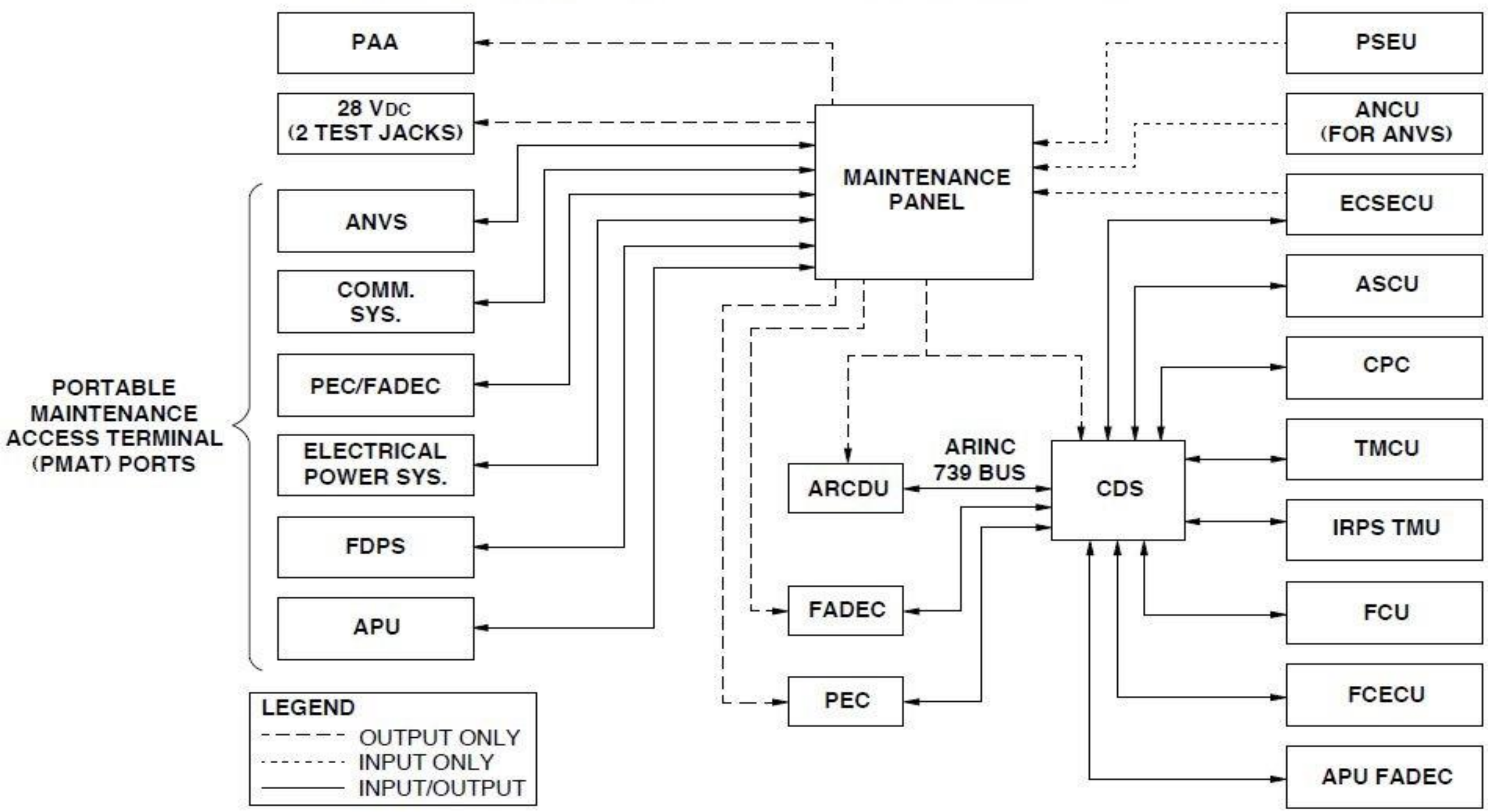


FIGURE - CDS MAINTENANCE PANEL BLOCK DIAGRAM

The maintenance panel interfaces with the components that follow:

Refer Figure - Maintenance Panel Interfacing With Other Components

- Full Authority Digital Electronic Control (FADEC)
- Propeller Electronic Controller (PEC)
- Environmental Control System Electronic Control Unit (ECSECU)
- Cabin Pressure Controller (CPC)
- Flight Control Electronic Control Unit (FCECU)
- Active Noise Control Unit (ANCU)
- Timer Monitor Control Unit (TMCU) and Timer Monitor Unit (TMU)
- Flight Control Electronic Control Unit (FCECU)
- Flap Control Unit (FCU)
- Proximity Sensor Electronic Unit (PSEU) and Anti−Skid Control Unit (ASCU)
- APU FADEC
- Passenger Address Amplifier (PAA).

Two test jacks on the maintenance panel can give a 28Vdc power supply while in the maintenance mode.

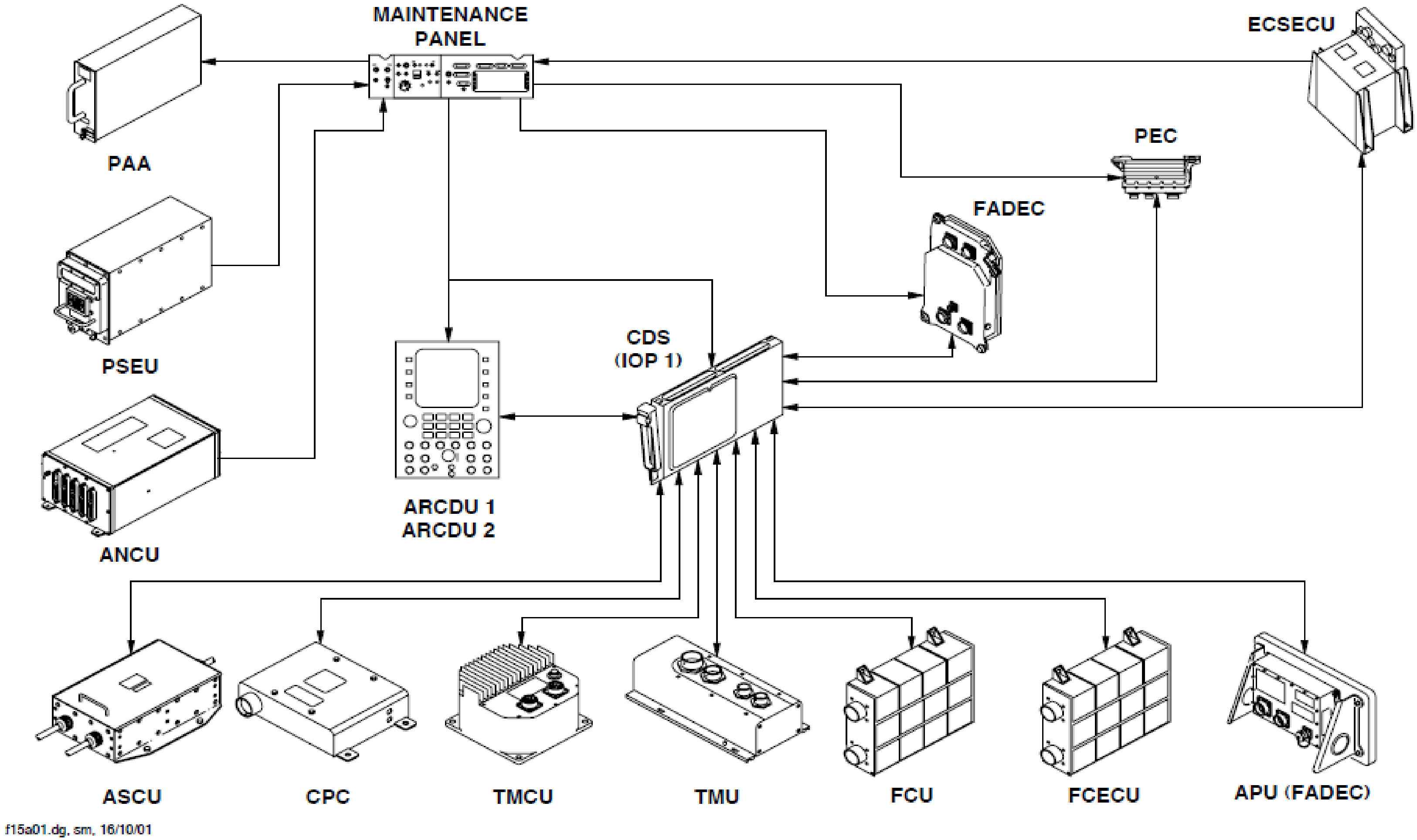


FIGURE - MAINTENANCE PANEL INTERFACING WITH OTHER COMPONENTS

Refer Figure - PMAT Block Diagram

The maintenance panel has six maintenance ports. They can be used to download data to external diagnostic locations (laptop computers). The systems and components that can download data are as follows:

- Active Noise Vibration Suppression (ANVS) System
- Communications System
- Propeller Electronic Controller (PEC) and Full Authority
- Digital Electronic Control (FADEC)
- Electrical Power System
- Flight Data Processing System (FDPS)
- Auxiliary Power Unit (APU)

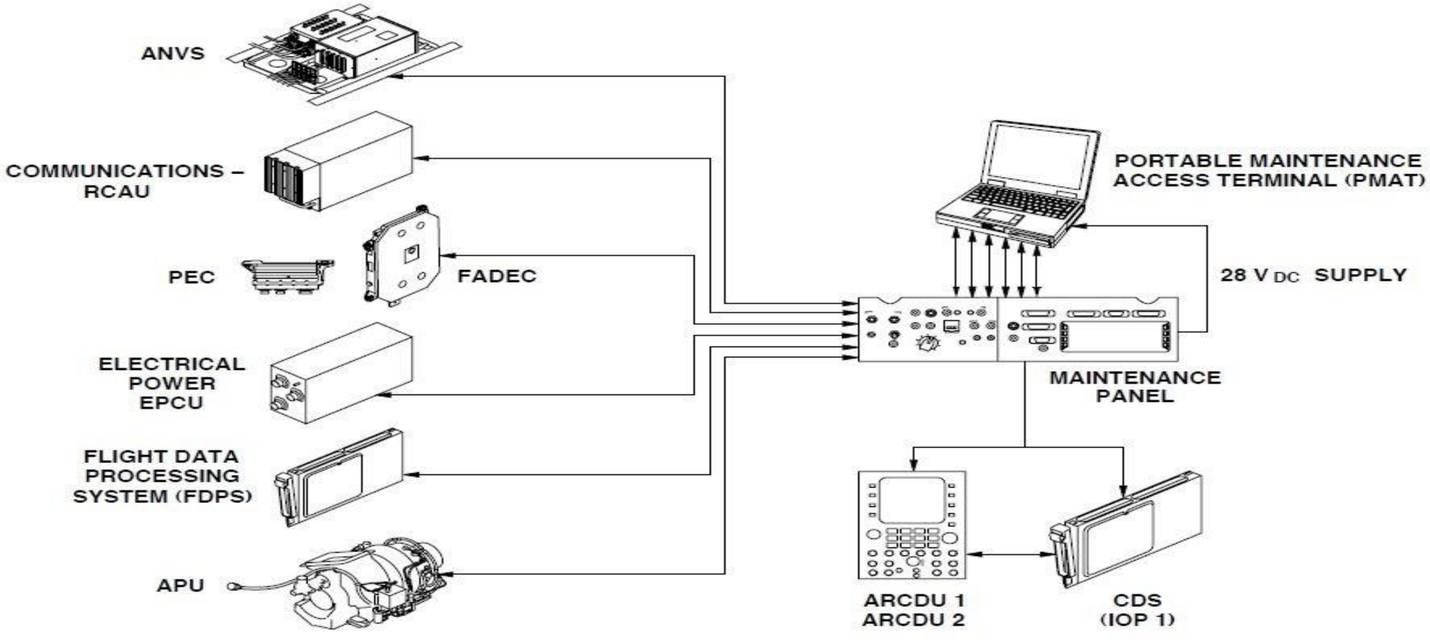


FIGURE - PMAT BLOCK DIAGRAM

Detailed Description

To use the maintenance panel, the CDS must be in the maintenance mode. This is the mode where systems give their fault data. The Audio Radio Control Display Unit (ARCDU) shows the fault codes. The two ARCDUs are installed on the flight compartment center−pedestal. The CDS is in the maintenance mode when the aircraft is in the conditions that follow:

- The calibrated airspeed (CAS) is less than 50 kts.
- The aircraft is on the ground (Weight on Wheels).
- The CDS GND MAINT toggle switch on the maintenance panel is set to the “up” position.
- The MAINT (maintenance) key on the ARCDU is pushed.

NOTE

Only one of the two ARCDUs is used at one time to show Maintenance pages. The selection of the CDS maintenance mode at one ARCDU prevents CDS operation from the other
ARCDU.

The keys that follow control the CDS and the CDS displays in the maintenance mode:

- MAINT key − Pushed to enter the maintenance mode and display the CDS main menu page
- NEXT key − Pushed to view the next CDS page
- PREV key − Pushed to view the previous CDS page
- DIM/RTN key − Pushed to return to the main page (when the ARCDU is operating in the maintenance mode).

Refer Figure - CDS ARCDU Selections

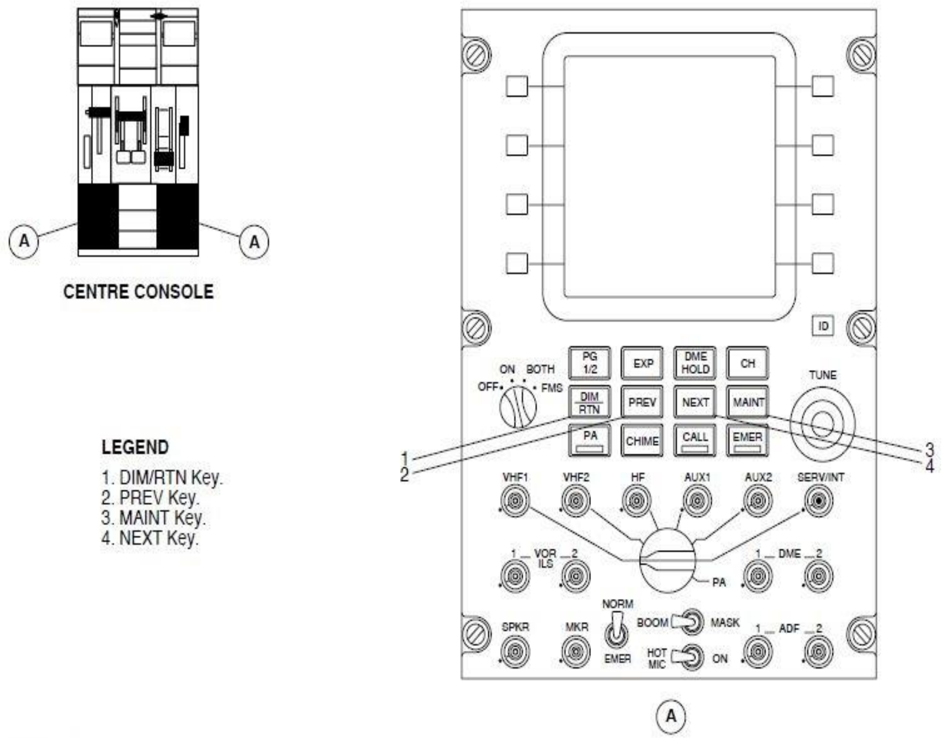


FIGURE - CDS ARCDU SELECTIONS

ENGINE MAINTENANCE PANEL

Refer Figure - Maintenance Panel and ARCDU – Engine Maintenance

The left part of the maintenance panel is the ENGINE MAINTENANCE Panel. This part of the maintenance panel is necessary for many calibration and functional tests. The powerplant and propeller systems interface with the ENGINE MAINTENANCE Panel. These systems are rigged/trimmed with aid of the switches on the ENGINE MAINTENANCE Panel.

Powerplant (74−10−00)

To check/test/rig the ignition system, the FADEC does an ignition test function in the maintenance mode. Set the MAINT DISC toggle switch to the MAINT DISC position. A maintenance discrete signal is sent to the Engine Cockpit Interface Units (ECIU) 1 and 2. This causes the MAINT DISC light emitting diode (LED) below the switch to come on.

Powerplant Rig Trim (61−20−00)

The MAINT DISC and RIG TRIM toggle switches are used to check the Condition Lever Angle (CLA) signal tolerance.

The CLA and the Power Lever Angle (PLA) are set on the ARCDU. The MAINT DISC toggle switch on the maintenance panel is set to the MAINT DISC position. The RIG TRIM toggle switch is then set in the RIG TRIM position. The ARCDU will show no faults at this time.

The MAINT DISC and RIG TRIM toggle switches are used to check/test/rig the FADEC.

The FADEC is rigged/trimmed to the flight idle position to the PLA Rotary Variable Differential Transformer (RVDT) input to the FADEC software. System tolerances then makes sure that the sensed PLA is the same between the two engines. The two FADECs can be done together if necessary.

The ARCDU is used to set the PLA and CLA. The MAINT DISC toggle switch is set to the MAINT DISC position. The RIG TRIM toggle switch is set to the RIG TRIM position. The ARCDU will show a trim value of 35 degrees for the two channels.

Propeller (61−20−00)

On the ground, a static propeller can be unfeathered while in the maintenance mode. The MAINT DISC toggle switch with the UNFEATHER #1 and #2 switches will unfeather the related propeller. These switches energize the auxiliary pump and the unfeather solenoid, which controls oil pressure to the servo valve. The PEC controls the servo valve output to drive the propeller towards fine pitch when:

- The propeller does not turn.
- The aircraft is in the maintenance mode.

The PEC can be trimmed with the maintenance panel. The aircraft must be in the maintenance mode. The PROPELLER PITCH TRIM page will show on the ARCDU in the maintenance mode.

The propeller will be in fine pitch. The opposite PLA is set to the DISC position. The opposite CLA is set to the FUEL OFF position. The CDS GND MAINT toggle switch on the maintenance panel is set in the “up” position. The MAINT DISC toggle switch is set to the MAINT DISC position. The RIG TRIM toggle switch is set to the RIG TRIM position. The ARCDU page will show COMPLETE, with a trim value of 16 or 17 degrees for the two channels.

The maintenance panel will erase fault codes from the Non−Volatile Memory (NVM) of the systems that follow:

- PEC
- FADEC

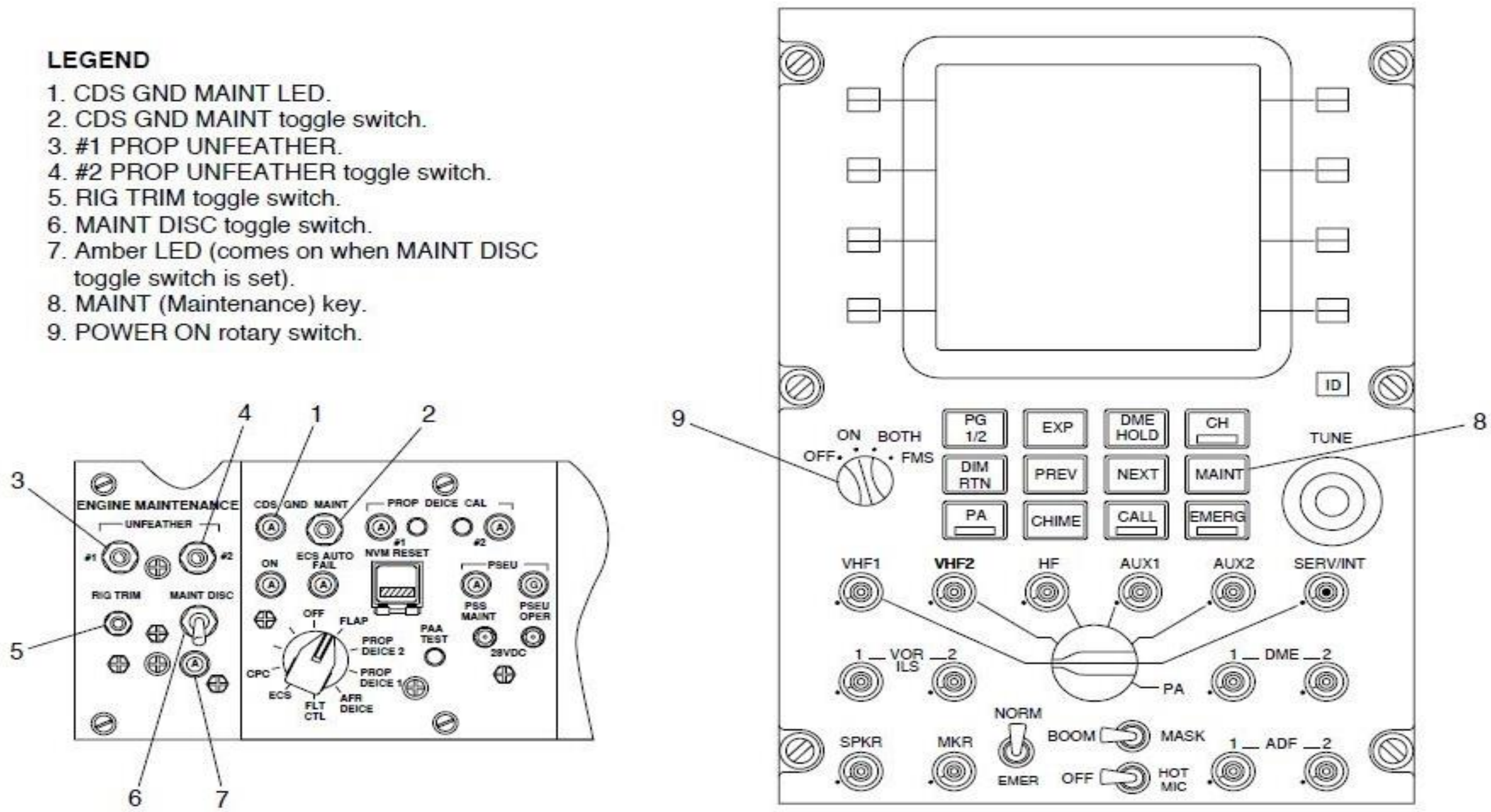


FIGURE - MAINTENANCE PANEL AND ARCDU − ENGINE MAINTENANCE

SYSTEMS MAINTENANCE PANEL

Refer Figure 45- 9 Maintenance Panel - **System Maintenance Panel**

The CDS GND MAINT toggle switch is set in the “up” position to give access to fault data through the CDS. This also causes the CDS GND MAINT LED to come on. The LED shows that the maintenance panel is energized.

The NVM RESET button on the maintenance panel is used to erase fault codes from the NVM of a system. The CDS and the maintenance panel must be energized to see the fault codes. The ARCDU shows fault codes when a system is set. The fault codes will be erased when the NVM RESET is pushed. The ARCDU will show a NO FAULT REPORTED message when the no fault codes are sensed.

NOTE

The fault codes must be erased after the correction of the fault is complete.

The rotary selector switch is used to erase fault codes for a specified system. When this switch is set to a position other than OFF, the amber LED located below the CDS GND MAINT LED comes on. This shows that a system has been set.

The MAINT key is pushed on the ARCDU to start the maintenance mode. A side key is pushed to see fault codes for a specified system.

Refer Figure - CDS Serviceable and No Fault Indication

Non−avionics systems are accessed through the OTHER SYSTEMS menu. This menu records the systems that follow:

- EMU (Engine Monitoring Unit)
- FCS (Flight Control System)
- ECS (Environmental Control System)
- TMCU1 (Timer Monitor Control Unit #1)
- ANTI−SKID
- ICE PROTECTION
- TMCU2 (Timer Monitor Control Unit #2)
- APU (Auxiliary Power Unit)
- FLAPS
- CABIN PRESSURE CONTROL
- AUXILIARY

NOTE

The AUXILIARY selection is not in use at this time. Selection of a system from this menu causes the CDS to send a discrete “transmit data” request to the respective unit. The EMU is the only selection that will not cause a CDS “transmit data” request.

Cabin Pressure Controller (CPC) (21−60−45)

The CPCS Failure Reporting page is shown when the Cabin Pressure Control side−key is set. The list of faults will show on one page.

The rotary selector switch to the CPC position. Push the NVM RESET button to erase all faults from the CPC memory.

Environmental Control System (ECS) (21−60−00)

The ECS AUTO FAIL light shows there is a failure recorded in the ECS system. The indication is only shown with weight on wheels (WOW). The Built−In−Test (BIT) function of the ECS Electronic Control Unit (ECSECU) will sense a system malfunction. The ECSECU fault codes are in the Non−Volatile Random Access Memory (NVRAM) of the ECSECU.

The CDS transmits a data request discrete when the ECS side−key is set. This causes data from the ECSECU to be sent to the CDS. The fault codes are then shown on the ARCDU. After the faults are corrected, the maintenance panel is used to erase the NVM fault code data.

The rotary selector switch is set to the ECS position. The NVM RESET button is pushed to erase all faults in the ECSECU.

Ice and Rain Protection System (IRPS) (30−61−00)

The two PROP DEICE CAL amber LEDs, labelled #1 and #2 come on when the propeller deice calibration is satisfactory. The push−button switches adjacent to the LEDs start the deice calibration.

NOTE

The propeller deice system calibration procedure can also be started from the Timer Monitor Control Units (TMCU). A switch is pushed on the top face of the two TMCUs to start the calibration procedure.

The TMCU monitors passive faults continuously for the propeller deice system. Fault data is transmitted by the TMCU on an ARINC 429 Digital Information Travel System (DITS) bus. This transmission occurs on request from the CDS.

The rotary selector switch is set to the PROP DEICE1 or PROP DEICE2 position. The NVM RESET button is pushed to erase all fault codes from the TMCU1 or TMCU2 memory.

A maximum of seven TMCU FAULTS pages can be seen on the ARCDU. The NEXT and PREV keys move from one page to another. The number of pages shown is a result of the number of malfunction conditions sensed. Each time an intermittent malfunction is sensed, it will be shown the same number of times on the TMCU fault pages.

The CDS can request Airframe Deice System fault codes from the IRPS Timer Monitor Unit (TMU) memory. Faults are given a priority level by the TMU. The most important faults are recorded first when they show on the ARCDU.

The rotary selector switch is set to the AFR DEICE position. The NVM RESET button is pushed to erase the fault codes from the IRPS TMU memory.

Flaps (27−51−00)

An asymmetry or an uncommanded fault can cause the FLAP POWER caution light to come on. This fault can only be erased with the NVM RESET button. The FCU controls, monitors, and gives caution indications of the flap control system.

The rotary selector switch is set to the FLAP position. The NVM RESET button is pushed to erase fault codes from the Flap Control Unit (FCU) memory.

Landing Gear (32−61−00)

The Proximity Sensor System (PSS) faults are recorded in the Proximity Sensor Electronic Unit (PSEU). When the PSEU contains faults, it sends an output signal to the maintenance panel. This signal causes the PSS MAINT LED to come on.

The PSEU is a component of the PSS that controls and monitors the operation of the landing gear components. A different discrete output from the PSEU causes the PSEU OPER LED to come on.

This shows power is received by all internal PSEU components. No PSEU faults are recorded in Present Faults when the PSEU OPER LED comes on.

Auxiliary Power Unit (APU) (49−70−00)

While the ARCDU is in the maintenance mode, the APU status and recorded APU faults can be shown. This occurs when the CDS transmits a data request from the APU FADEC.

Public Address Amplifier (23−31−00)

The selection of the Public Address Amplifier press−to−test switch (labelled PAA TEST) supplies a PAA discrete input. This causes a continuous tone to make sure of loudspeaker operation in the cabin and flight compartment. The Public Address Amplifier (PAA) interfaces with the cabin speaker system. It is controlled by the ARCDU, the forward attendant headset, and the aft attendant headset.

Test Jacks

The test jacks can be used for maintenance where a 28Vdc supply is necessary. The location of the test jacks is immediately below the PSEU LEDs. The 28 Vdc Left Secondary Feeder bus is the power supply for the two 28Vdc test jacks on the maintenance panel.

NOTE

The test jacks are energized when the PWR ON toggle switch is set in the “down” position.

Refer Figure - Maintenance Panel – System Download Panel

The PWR ON toggle switch energizes the two 28 Vdc test jacks on the System Maintenance panel. The switch position is on the left side of the SYSTEM DOWNLOAD Panel. When the PWR ON toggle switch is set in the “down” position, 28 Vdc is supplied to the test jacks. The LED below the PWR ON toggle

switch comes on to show that power is supplied to the test jacks.

The LED TEST is not functional at this time.

The Portable Multipurpose Access Terminal (PMAT) ports are on the right side of the SYSTEM DOWNLOAD panel. There are six ports. The PMAT ports (J1 to J5) are for future use. A PMAT for this aircraft is not available at this time. Each port gives the PMAT access to specified systems as follows:

J1 AVIONIC RS422/A429 is the PMAT port for the Communications System. This port lets the PMAT interface with the Remote Control Audio Unit (RCAU).

J3 AVIONIC RS422/A429 is the PMAT port for the Electrical Power System. The PMAT can interface with the Electrical Power Control Unit (EPCU) through this port.

J2 AVIONIC RS422/A429 is the PMAT port to transfer data from the PEC or the FADEC to the PMAT.

J4 AVIONIC A429 is the PMAT port for the Flight Data Processing System (FDPS). Flight data can be downloaded through this port. J5 APU RS422 is the PMAT port for the APU system. Data can be downloaded from the APU FADEC NVM. The download starts on the command from the PMAT.

The ANVS SYS RS232 maintenance port can connect to a personal computer (maintenance terminal). It interfaces with the ANVS Controller to troubleshoot and calibrate. (23−35−00)

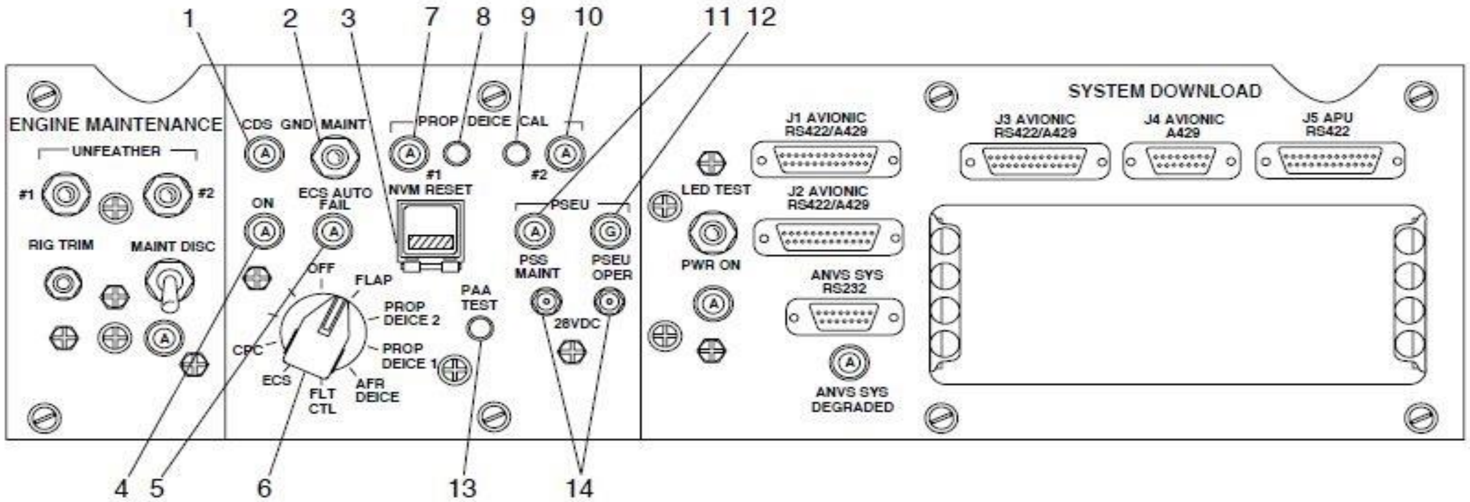
Active Noise Vibration Suppression (ANVS) (MTM 23−35−00)

The ANVS system has an amber LED labelled ANVS SYS DEGRADED on the maintenance panel. If the number of failures is more than the degraded limit, the LED comes on. This will show that the ANVS performance is degraded. The degraded limit is set to two failed channels. ANVS system faults are corrected through an external maintenance terminal through the ANVS SYS RS232 maintenance port.

The maintenance panel has an opening for the installation of a Digital Passenger Announcement System (DPAS) Unit. The opening for the DPAS is on the SYSTEM DOWNLOAD panel adjacent to the ANVS SYS RS232 maintenance port. The DPAS is a part of the Passenger Entertainment System (PES).

LEGEND

1. CDS GND MAINT light.
2. CDS GND MAINT toggle switch.
3. NVM RESET (Guarded) push–button.
4. System selector light.
5. ECS AUTO FAIL light.
6. System selector rotary switch.
7. #1 PROP DEICE CAL light.
8. #1 PROP DEICE CAL push–button.
9. #2 PROP DEICE CAL push–button.
10. #2 PROP DEICE CAL light.
11. PSS MAINT light.
12. PSEU OPER light.
13. PAA TEST push–button.
14. 28 V_{DC} test jacks.



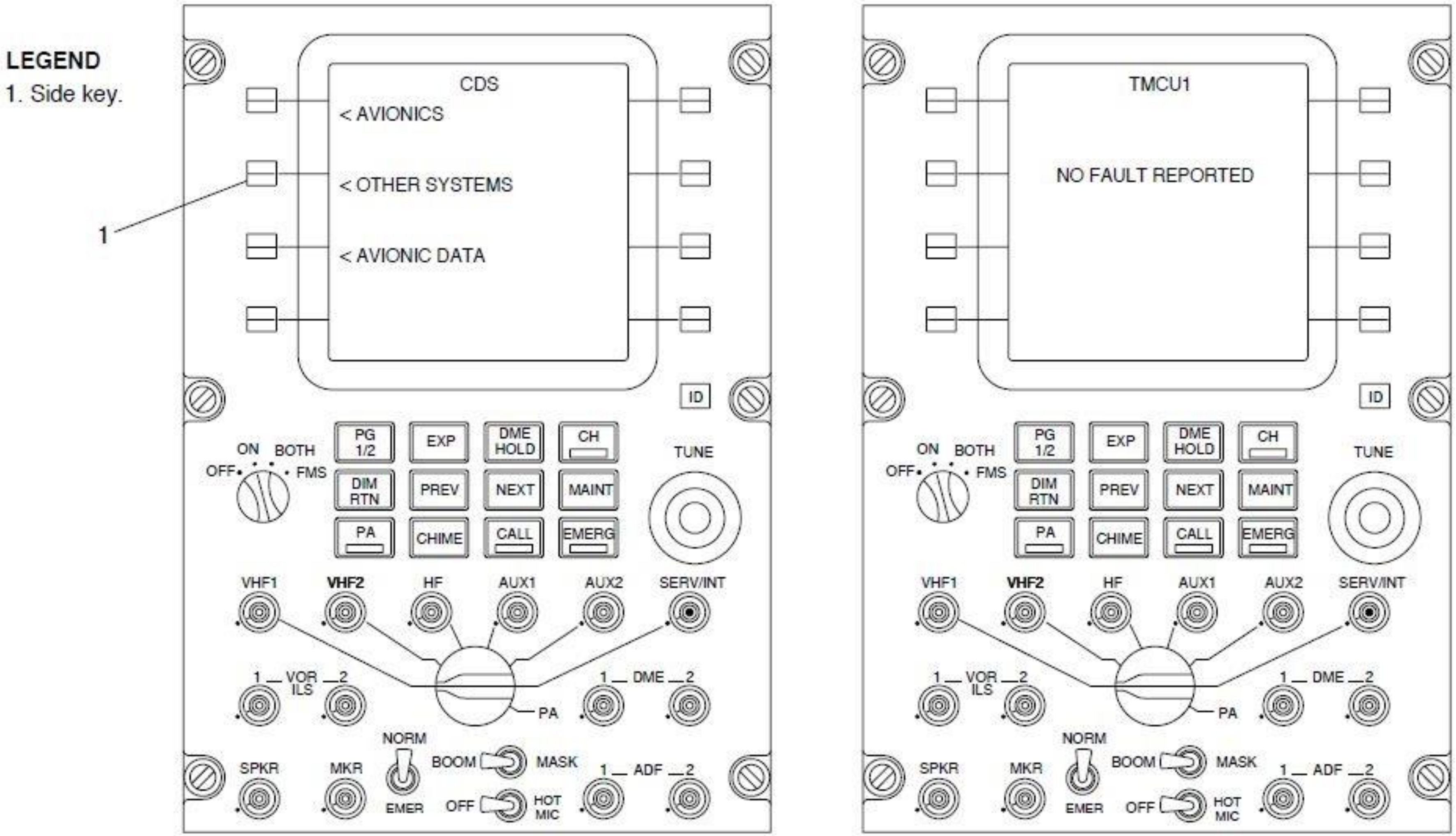


FIGURE - CDS SERVICEABLE AND NO FAULT INDICATION

LEGEND

1. J1 AVIONIC RS422/A429 Port (Communication System).
2. J2 AVIONIC RS422/A429 Port (PEC/FADEC).
3. J3 AVIONIC RS422/A429 Port (Electrical Power System).
4. J4 AVIONIC A429 Port (FDPS).
5. J5 APU RS422 Port (APU).
6. PWR ON Toggle Switch.
7. PWR ON light.
8. ANVS SYS DEGRADED light.
9. ANVS SYS RS232 Port (ANVS).
10. Digital Passenger Announcement System (DPAS) location.

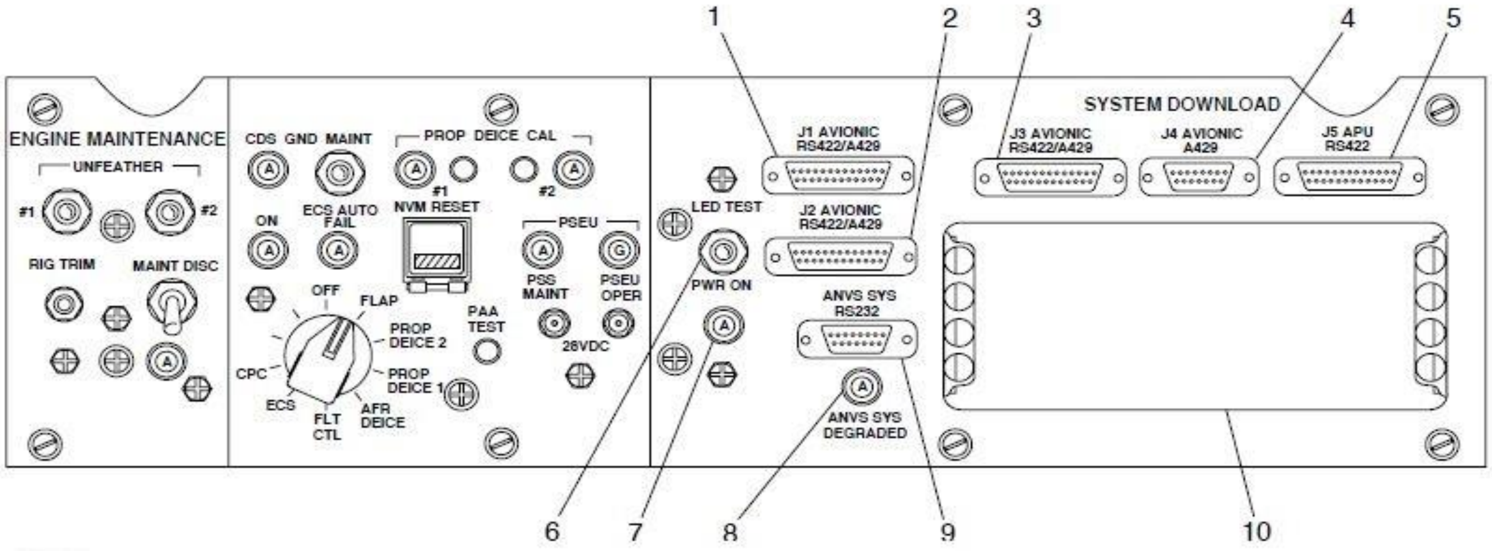


FIGURE - MAINTENANCE PANEL − SYSTEM DOWNLOAD PANEL